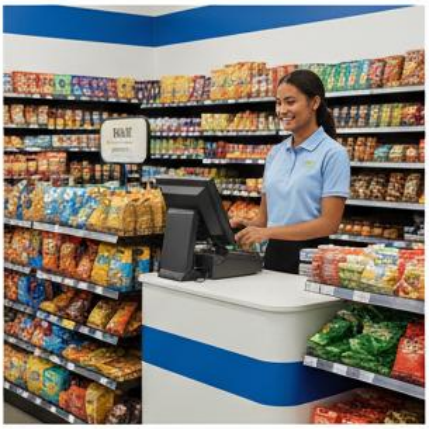
PRIMESTOP



Your Everyday, Elevated

PRIMESTOP CONVENIENCE

ADVANCED DATA MINING TO IMPROVE CUSTOMER CONVERSION

PRIMESTOP CONVENIENCE: TURNING FUEL CUSTOMERS INTO IN-STORE SHOPPERS

In the Mid-Atlantic fuel retail and convenience store market, PrimeStop has built a strong presence across Maryland, Virginia, and Washington, D.C. The brand is known for its clean locations, reliable fuel service, and a growing portfolio of stores offering food service, car wash, and enhanced convenience retail.

By 2023, PrimeStop had expanded aggressively into full-service convenience formats while facing increasing competition from national chains such as Wawa, Sheetz, and Royal Farms. Despite the brand's investments in elevated store experiences, loyalty programs, and targeted marketing, leadership faced a persistent challenge:

A large share of customers still behave as fuel-only users and do not regularly purchase from the convenience store.

PrimeStop's leadership believes that harnessing customer and store-level data more effectively could be the key to improving conversion from fuel-only to fuel + convenience shoppers, thereby increasing basket size and overall profitability.

In a strategy meeting at PrimeStop's headquarters, two executives, Maria Lopez, the Chief Marketing Officer, and Derrick Brown, the Director of Analytics and Customer Insights, sat down to review recent performance.

"We've got solid fuel traffic, but our convenience conversion rate isn't where it needs to be," Maria began, pointing to a dashboard on the screen. "We need to understand which customers are most likely to start shopping inside the store and what's holding others back."

Derrick nodded. "We have rich data on customer demographics, fuel behavior, store amenities, loyalty enrollment, promotions, and competition. But right now, we're underutilizing it. We need to go beyond descriptive reporting and move into predictive modeling."

Maria leaned forward. "Exactly. If we can identify the factors that drive conversion, we can tailor incentives, refine our loyalty program, and focus investments on the right stores and segments."

Derrick continued, "This is a classic classification problem. We can use advanced data mining methods to predict whether a customer will convert from fuel-only to convenience shopper. For those who convert, we can also model their in-store engagement—how often they visit and how much they tend to spend."

Maria smiled. "Let's do it. PrimeStop has always focused on convenience and customer experience. It's time we let the data guide us to the next level."

Approach and Expectations

As data consultants for **PrimeStop Convenience**, your task is to conduct an exploratory analysis and develop **classification and regression models**. The aim is to enhance PrimeStop's ability to:

- Predict which fuel customers are likely to become convenience shoppers, and
- Understand how engaged those converted customers will be with the in-store offering.

Specifically, PrimeStop is interested in understanding what customer, behavioral, and store-level attributes contribute to **conversion** and **downstream convenience behavior**.

Key Questions:

- Which customer and store attributes most significantly drive conversion from fuel-only to convenience shopper at PrimeStop?
- Can we develop classification models to predict the likelihood that a given customer will convert (start using the convenience store) within a short time window?
- For customers who convert, can we build regression models to predict their **total convenience visits** and/or **average convenience purchase amount** over a 30-day period?
- How do different models (Logistic Regression, LDA, QDA, KNN, Tree based models, Ensemble methods) compare in terms of performance metrics such as **Confusion Matrix, F1 Score, AUC-ROC, and ROC curves**?
  And for regression models, what are the **RMSE** and **R^2** values?

Data Dictionary

Variable Name	Description	Type	Possible Values
customer_id	Unique identifier for each customer	Numeric	1, 2, 3...
age_group	Age category of the customer	Categorical	18-25, 26-35, 36-45, 46-55, 56-65, 65+
gender	Gender of the customer	Categorical	Male, Female
income_level	Customer income bracket	Categorical	Low, Middle, High
occupation	Occupation category	Categorical	Professional, Service, Student, Retired, Blue Collar, Healthcare, Education
household_size	Number of people in the household	Numeric	1–6
commute_distance_miles	Commute distance in miles	Numeric	1–40
store_id	ID of primary store visited	Numeric	1001–1500
state	State of primary store	Categorical	MD, VA, DC
county	County or DC region	Categorical	Prince Georges, Fairfax, Northwest, etc.
location_type	Store location type	Categorical	Urban, Suburban, Highway, Rural
store_type	Store format	Categorical	Fuel Only, Convenience Store, Full Service + Car Wash
has_food_service	Store has food service	Binary	0 = No, 1 = Yes
has_car_wash	Store has car wash	Binary	0 = No, 1 = Yes
has_loyalty_program	Store offers loyalty program	Binary	0 = No, 1 = Yes
competitor_within_1mile	Competitor within 1 mile	Binary	0 = No, 1 = Yes
store_age_years	Age of store in years	Numeric	1–25
parking_spots	Number of parking spaces	Numeric	~8–60
days_since_first_visit	Days since customer's first visit	Numeric	1–30

initial_visit_day_of_week	Day of first visit	Categorical	Monday–Sunday
initial_visit_time_of_day	Time of first visit	Categorical	Morning, Afternoon, Evening, Night
primary_payment_method	Most used payment method	Categorical	Credit Card, Debit Card, Cash, Mobile Pay
enrolled_in_loyalty	Enrolled in loyalty program	Binary	0 = No, 1 = Yes
visit_frequency	Fuel visit frequency	Categorical	Daily, Weekly, Bi-weekly, Monthly
total_fuel_purchases_30days	Fuel purchases in past 30 days	Numeric	1–8
avg_fuel_amount_gallons	Average gallons per fuel visit	Numeric	6–22
preferred_visit_time	Preferred store visit time	Categorical	Morning, Afternoon, Evening
shops_seasonally	Shops more during seasonal periods	Binary	0 = No, 1 = Yes
used_promotion_before	Has redeemed a promotion before	Binary	0 = No, 1 = Yes
feels_safe_at_location	Customer feels safe at store	Binary	0 = No, 1 = Yes
satisfied_with_service	Customer satisfied with service	Binary	0 = No, 1 = Yes
converted	**Target variable** — converted to convenience shopper	Binary	0 = No, 1 = Yes
total_convenience_visits	Convenience visits in past 30 days	Numeric	0–25
avg_convenience_purchase_amount	Average convenience spend per visit	Numeric	0–20
most_purchased_category	Most purchased in-store category	Categorical	None, Beverages, Snacks, Food Service, Grocery